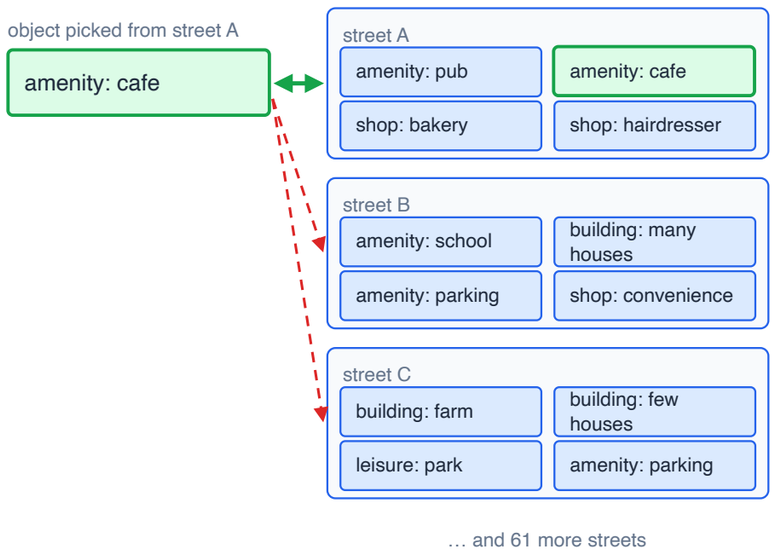
Stage 1: how CityFM learns what is found together, and how we test it

Everything on the map is read as its type only (e.g. *amenity: cafe*), with no names. A street is described by the list of things near it.

the picked object · removed from its street · made more similar: its own street · made less similar: the other 63 streets and objects

1 Training

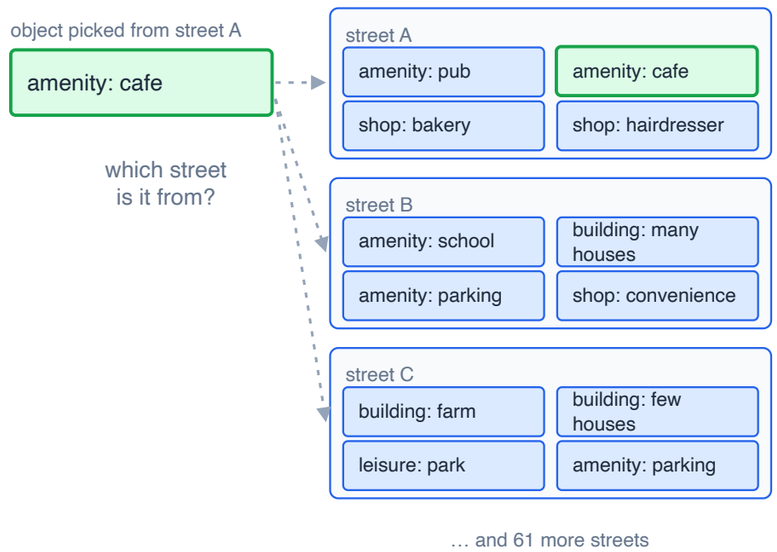
Take 64 streets and pick one object from each. The model is adjusted so that each object is more similar to its own street than to the other 63 streets and the other 63 objects. Repeated for about 2,500 quizzes.



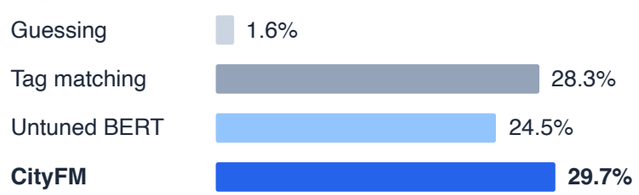
It can win in two ways
- spot the identical item on the street (a shortcut)
- learn which kinds of place are found together (the goal)

2 Test 1 · Training-style

The finished model plays the same game. The object **stays** on its street, so spotting the identical item is enough.



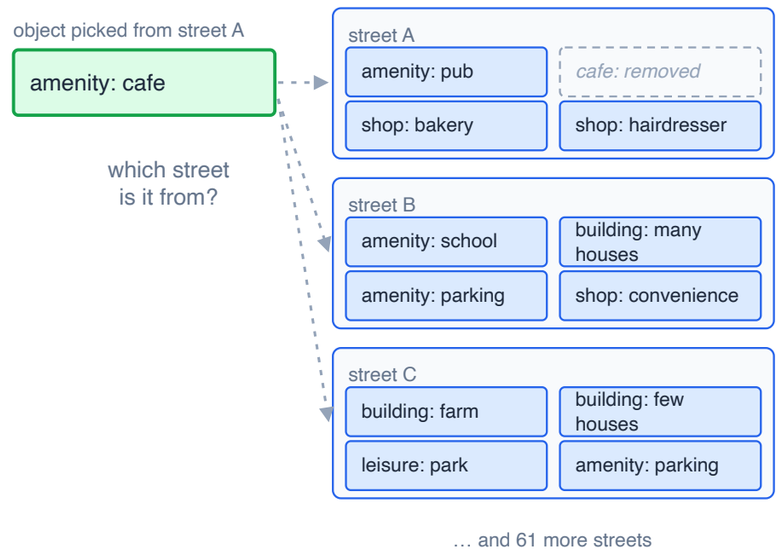
Right street picked first, out of 64



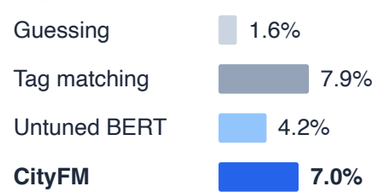
Easy for everyone: the answer is sitting in the list.

3 Test 2 · POI held out

The object is **removed** from its street first. The model must judge from the other things there: a pub, a bakery and a hairdresser suggest a café-like street.



Right street picked first, out of 64



Much harder. CityFM beats untuned BERT (and has the right street in its top 5 in 23% of quizzes).

What it shows. Training has taught CityFM something: it beats the untuned language model in both tests. But with the object held out it is no better than tag matching, so it may lean on spotting repeats. **Next:** hold the object out during training too, so the shortcut is not available.

Guessing: 1 in 64. **Tag matching**: no model; pick the street with the most items of exactly the same type. **Untuned BERT**: the language model before CityFM's training. Same scale in both charts.